

# IT5106 - Software Development

## Interim report – 2025

**Project Title:**

Tax, Garbage & Tender Details Management System for Puttalam  
Municipal Council

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# **ABSTRACT**

This project focuses on developing a web-based Digital Municipal Service Automation System for improving the efficiency of municipal council operations. The system aims to address issues in existing manual processes such as tax management, garbage collection, and tender handling.

The proposed solution provides a centralized platform for managing tax records, garbage scheduling, and tender processes, while ensuring transparency and reducing manual errors. The system is developed using Spring Boot for backend services, Angular for frontend development, and MySQL for database management.

Currently, system analysis and design phases have been completed, and initial development of core modules is in progress. The project is expected to significantly improve operational efficiency and service delivery within the municipal council.

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# **CHAPTER 1: INTRODUCTION**

## **1.1 Background**

Municipal councils play a vital role in providing essential public services such as tax collection, garbage management, and tender processing. However, many of these operations are still handled manually, leading to inefficiencies and delays.

## **1.2 Problem Statement**

The current system used in municipal councils is largely manual and paper-based. This leads to several issues such as time wastage, lack of transparency, data inconsistency, and difficulty in managing records.

## **1.3 Motivation**

The motivation for this project arises from the need to modernize municipal operations. Manual systems result in delays in tax processing, poor complaint handling, and inefficient garbage collection scheduling.

## **1.4 Project Overview**

This project proposes a web-based system that automates tax management, garbage management, and tender processes. It provides a centralized system for managing data, improving accuracy, and enhancing communication between the council and the public.

# CHAPTER 2: ANALYSIS

## 2.1 Existing System

Currently, municipal council operations are handled manually using paper-based records and basic tools like Excel.

Problems in Existing System:

- Time-consuming manual processes
- Lack of transparency in complaint handling
- Poor record management
- Risk of data loss
- Inefficient garbage scheduling
- Manual tender processing

## 2.2 Functional Requirements

The system should be able to:

Tax Management

- Create tax records by officer
- Automatically calculate penalty after due date
- Display tax status in user dashboard
- Mark tax as paid manually by officer

Garbage Management

- Register garbage vehicles
- Assign vehicles to schedules
- Upload garbage weight
- Automatically generate garbage fee based on weight

Tender Management

- Publish tenders by officer
- Submit bids by business users
- Evaluate bids after closing date

- Award tenders and update dashboard

#### User Management

- User registration and login
- Role-based access (admin, staff, citizens)

#### Reporting

- Generate reports
- Display dashboards

## 2.3 Non-Functional Requirements

- Security (user authentication)
- Performance (fast processing)
- Usability (easy interface)
- Reliability (accurate data handling)
- Scalability (future expansion possible)

## 2.4 Literature Review

With the rapid advancement of information technology, many governments and municipal councils around the world have started adopting digital systems to improve public service delivery. These systems, commonly referred to as e-government systems, aim to replace traditional manual processes with automated and centralized platforms.

E-government systems are widely used to manage services such as tax collection, citizen registration, complaint handling, and public resource management. According to various studies, implementing such systems significantly improves efficiency, reduces operational costs, and enhances transparency between government authorities and citizens.

Municipal management systems are a specific category of e-government solutions that focus on local government operations. These systems typically include modules for property tax management, waste collection scheduling, and infrastructure maintenance. Research shows that digital municipal systems help reduce paperwork, minimize human errors, and improve record management.

In addition, smart city initiatives have introduced more advanced solutions that integrate technologies such as real-time monitoring, data

analytics, and automated workflows. These systems allow authorities to track garbage collection, monitor service delivery, and make data-driven decisions.

Several existing web-based municipal systems also provide centralized dashboards for administrators, enabling them to view reports, manage operations, and monitor performance in real time. These systems highlight the importance of integrating multiple services into a single platform for better coordination and efficiency.

Based on the above studies and existing systems, it is evident that developing a web-based municipal service management system can significantly improve operational efficiency, transparency, and service quality. The proposed system in this project is designed by considering these concepts and aims to provide a practical solution tailored to the needs of a municipal council.



# **CHAPTER 3: DESIGN OF SOLUTION**

## **3.1 Proposed Solution**

The proposed system is a web-based application that integrates tax, garbage, and tender management into a single platform. It uses a client-server architecture with Angular frontend and Spring Boot backend.

## **3.2 Alternative Solutions**

Alternative approaches include:

- Fully manual system (inefficient)
- Standalone desktop applications (limited access)

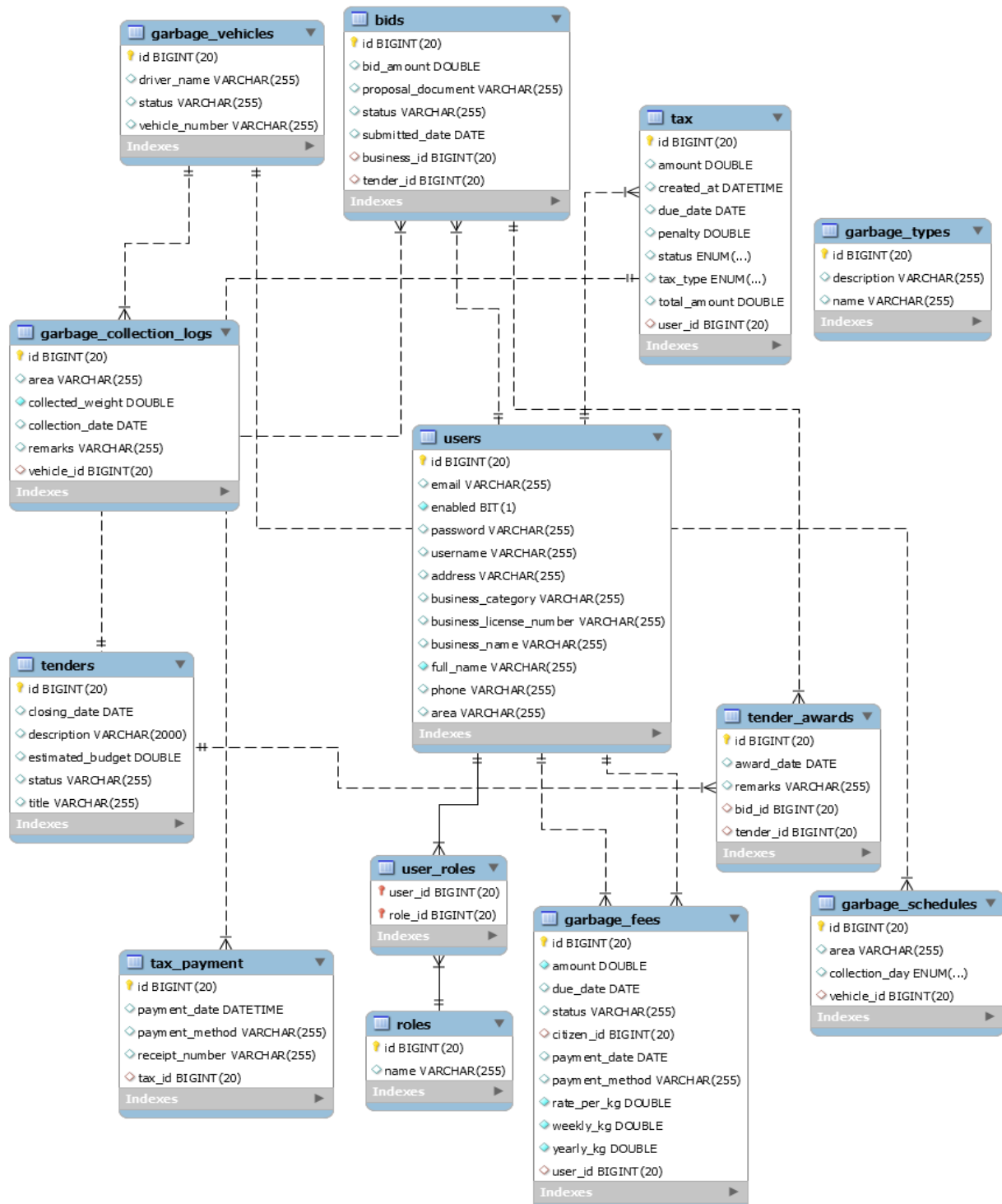
The selected web-based system is preferred due to accessibility, scalability, and centralized data management.

## **3.3 System Architecture**

- Frontend: Angular
- Backend: Spring Boot
- Database: MySQL

## 3.4 Diagrams

- ER Diagram



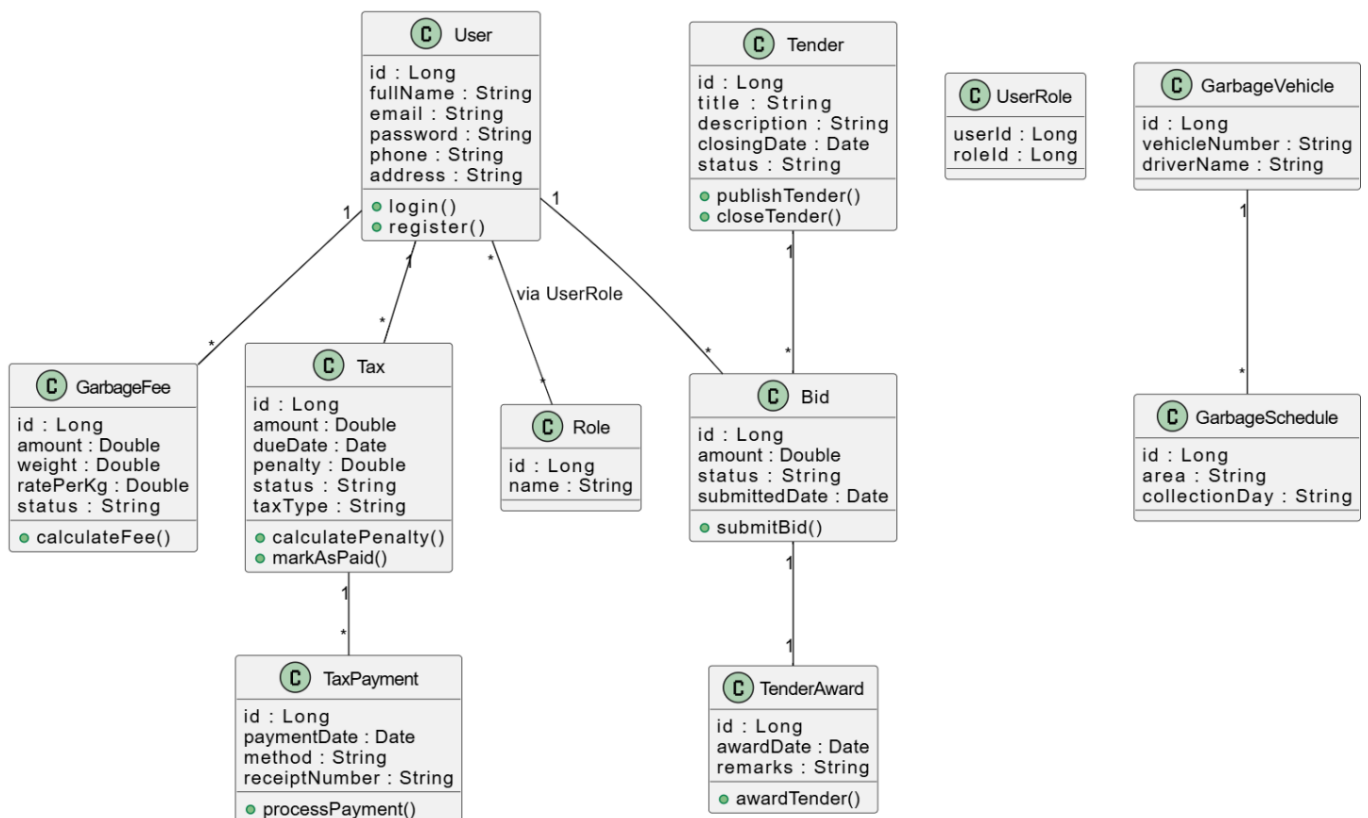
The ER diagram represents the database structure of the system, showing relationships between entities such as users, tax records, garbage management, and tender operations.

- Use Case Diagram



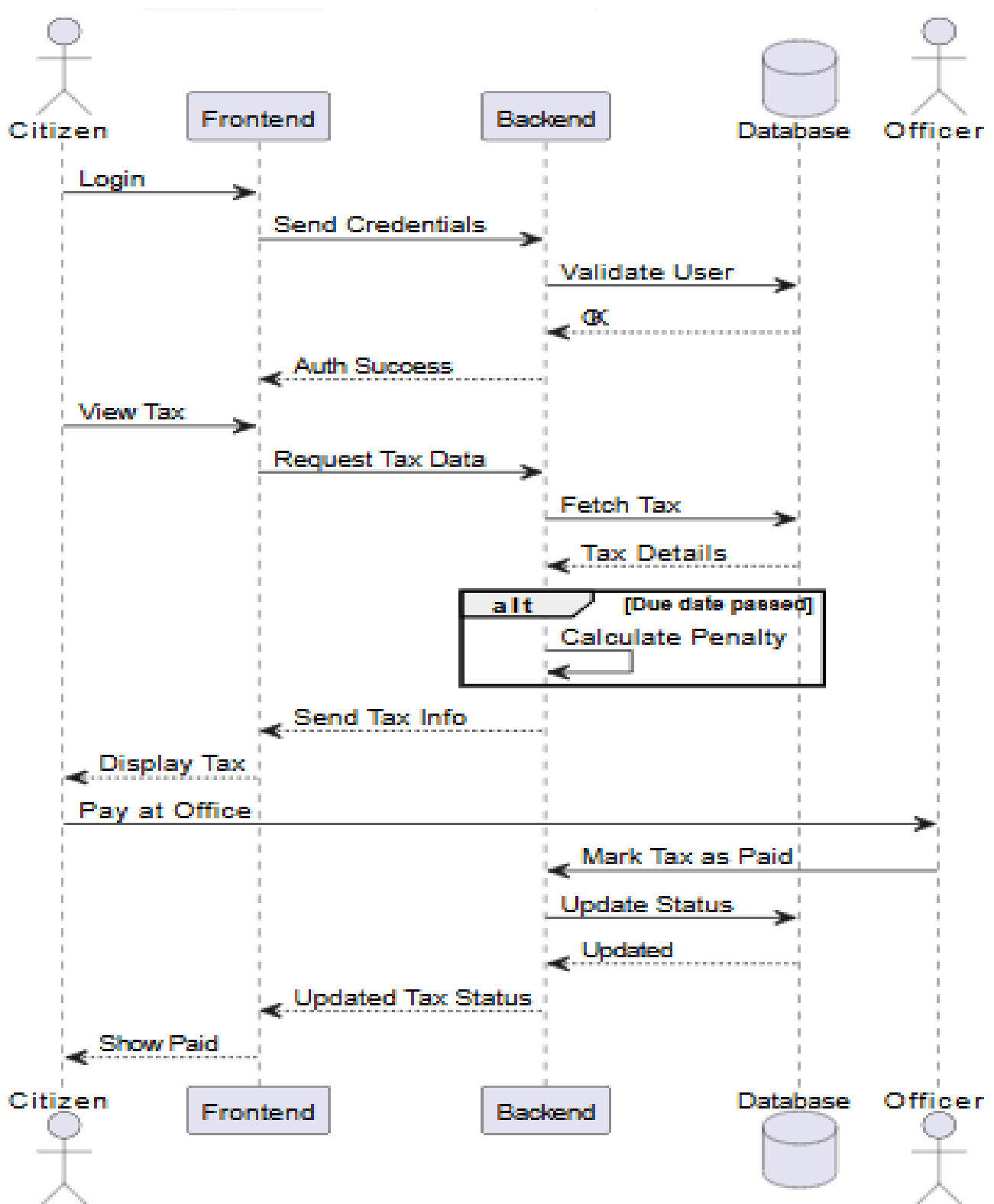
The use case diagram represents the interaction between system users including Admin, Officer, Citizen, and Business users. It illustrates key functionalities such as tax management, tender processing, garbage management, and user handling. Certain operations such as penalty calculation and garbage fee generation are automatically triggered within the system using included relationships.

- Class Diagram

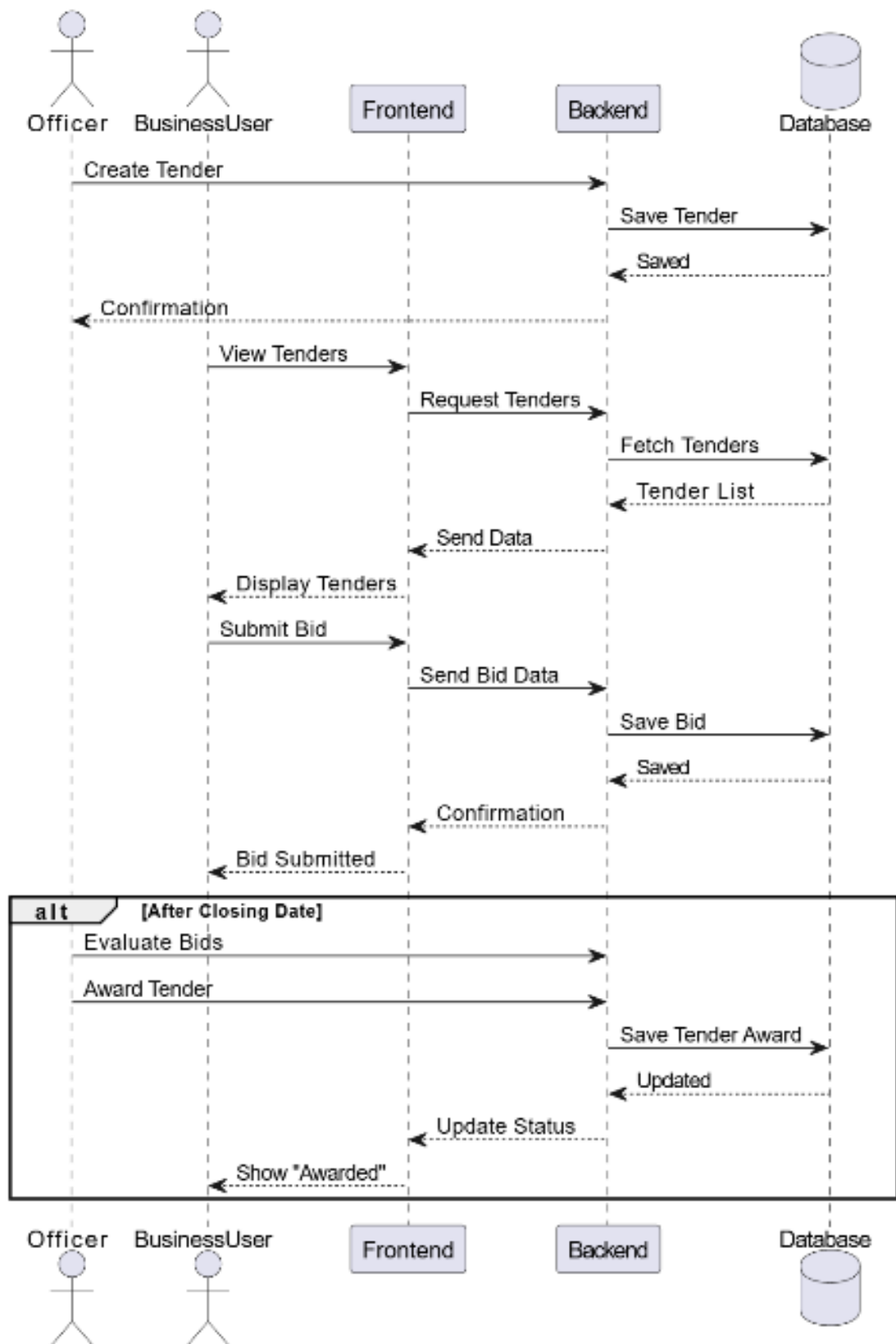


The class diagram illustrates the system structure in terms of classes, attributes, and methods, derived from the database and application logic.

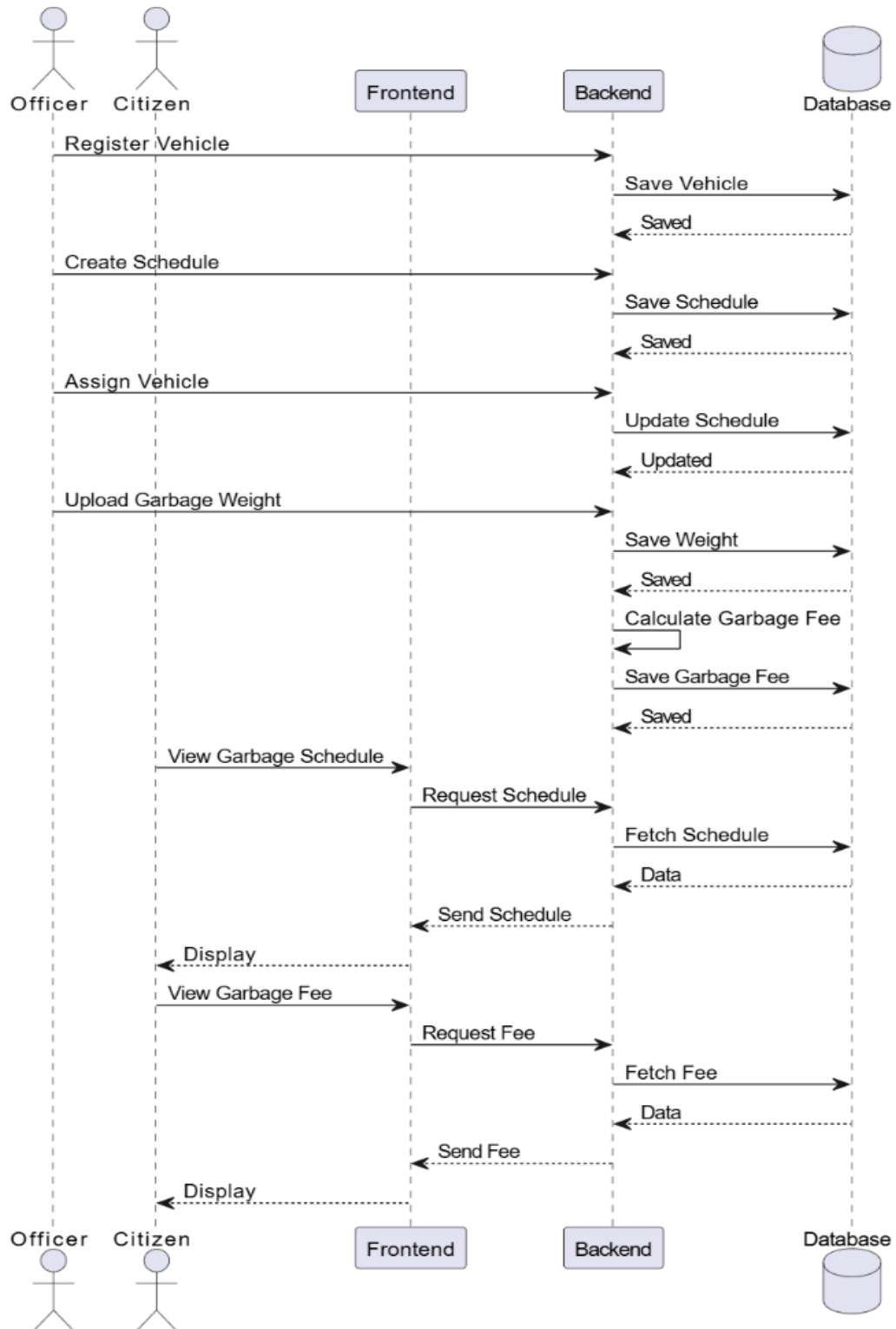
- Sequence Diagram (Tax flow)



- Sequence Diagram (Tender flow)



- Sequence Diagram (Garbage flow)



This sequence diagram shows the interaction between system components during the execution of the [tax/tender/garbage] process.

### **3.5 Database Design**

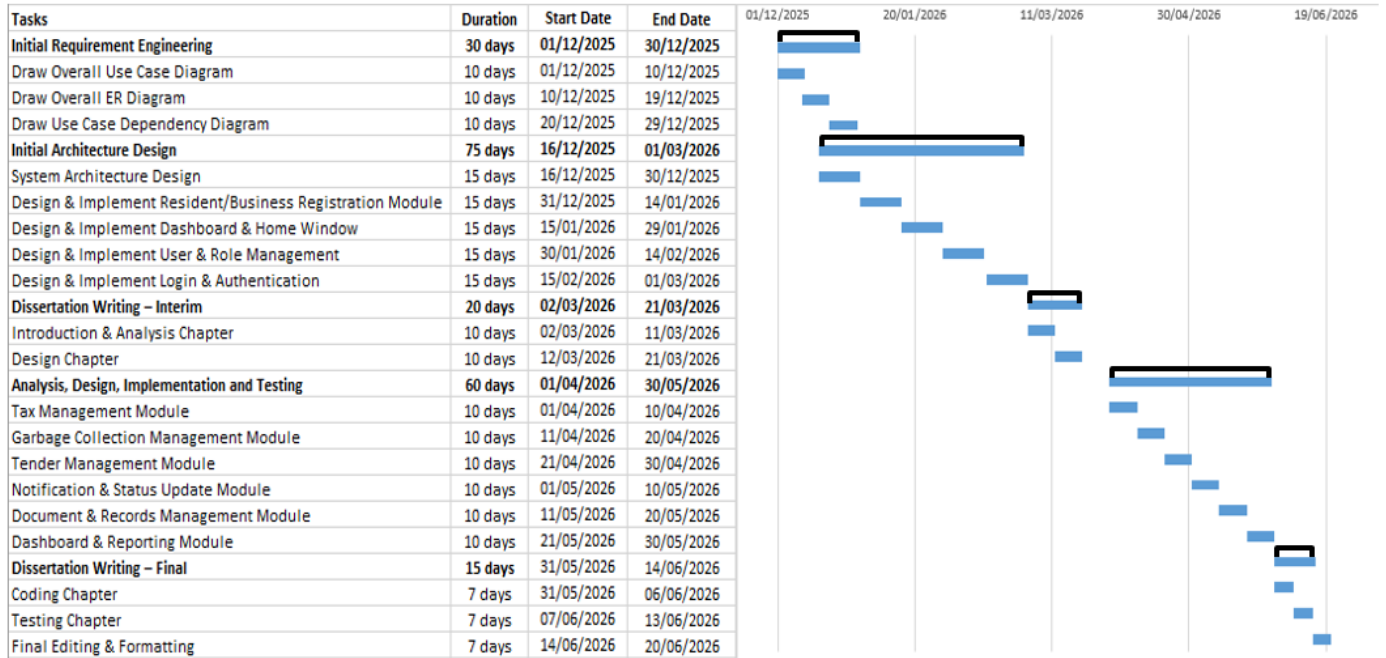
The database consists of multiple related entities including Users, Roles, Tax, Tax Payments, Garbage Vehicles, Garbage Schedules, Garbage Fees, Tenders, Bids, and Tender Awards. These entities are interconnected to support system operations such as tax processing, garbage fee calculation, and tender management.

### **3.6 Tools & Technologies**

- Java (Spring Boot)
- Angular
- MySQL
- Postman
- GitHub



# Gantt Chart

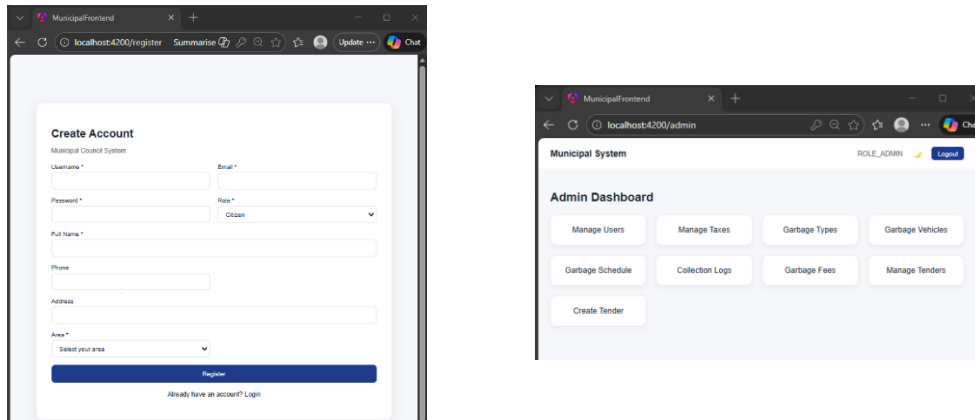


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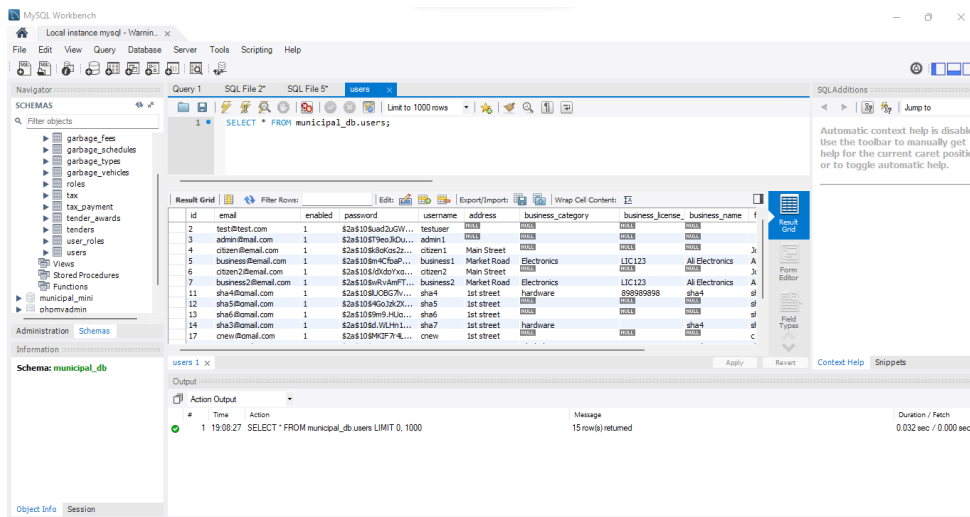
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# APPENDICES

## Appendix A: System Screenshots



## Appendix B: Database Design



## Appendix C: API Testing Results

